A Converse Estimate for the Schlumprecht-Space Szlenk Radius

Automated Math Discovery Smoke Test

June 14, 2026

Source and Open Problem

This packet concerns Proposition 6 and the open problem immediately following it in:

> Tomasz Kochanek and Marek Miarka, *When is the Szlenk derivation of a dual unit ball another ball?*, arXiv:2409.05516 [1].

The source location is page 9 of the paper.

$$\|x_0 + \frac{\varepsilon}{2}e_n\| = \max\left\{r, \frac{\varepsilon}{2}, \frac{r+\frac{\varepsilon}{2}}{\phi(2)}\right\} < 1,$$

which shows that $x_0 \in s_\varepsilon B_{\mathcal{S}}$ and completes the proof. □

Open problem. We do not know if the inequality converse to (6) holds true.

4. Derivations of Baernstein's space

Let $\mathcal{S}$ be Schlumprecht's space [2]. Its norm is given implicitly by

$$\|x\|_{\mathcal{S}} = \max\left\{\|x\|_\infty, \sup_{E_1<\cdots<E_n} \frac{1}{\varphi(n)}\sum_{i=1}^{n}\|E_i x\|_{\mathcal{S}}\right\}, \qquad \varphi(n) = \log_2(n+1).$$

The case $n = 1$ is redundant, since $\varphi(1) = 1$, so we shall use only $n \geq 2$ in recursive estimates.

Kochanek and Miarka proved

$$r_{\mathcal{S}^*}(\varepsilon) \leq \inf_{n\in\mathbb{N}} \frac{\varphi(n+1) - \varepsilon/2}{\varphi(n)}, \qquad 0 < \varepsilon < 2, \tag{1}$$

and asked whether the converse inequality also holds.

Proof Intuition

The missing direction is a far-coordinate perturbation argument. If a vector $x \in \mathcal{S}$ has norm below the candidate radius, then one wants to add and subtract a sufficiently far basis vector αe_N while staying in the unit ball. The two perturbed vectors converge weakly back to x and remain separated by 2α, which places x in the Szlenk derivation.

The only delicate point is that the scalar majorant must be local: when a restriction F is estimated recursively, the sub-restrictions have to be controlled relative to the majorant of F itself. The needed bound has the form

$$\sum_i \rho(E_i) \leq \varphi(k)\rho(F).$$

The key observation in this revision is that the needed strengthened scalar estimate follows from a simple product inequality for the logarithmic increments of φ.

1 A Logarithmic Product Lemma

Put
$$B_i = \varphi(i+1), \qquad \Delta_i = \varphi(i+1) - \varphi(i) \qquad (i \geq 1).$$

Lemma 1 (monotonicity)**.** *The function*
$$h(u) = \frac{u \log u}{u-1} \qquad (u > 1)$$
is increasing.

Proof. Differentiation gives
$$h'(u) = \frac{u - 1 - \log u}{(u-1)^2} \geq 0,$$
because $\log u \leq u - 1$. □

Lemma 2 (product lemma)**.** *For all $i, j \geq 1$, if $\ell = (i+1)(j+1) - 1$, then*
$$B_\ell \leq B_i B_j, \qquad \Delta_\ell \leq \Delta_i \Delta_j.$$

Proof. It is enough to prove, for $p, q \geq 1$,
$$\log_2(pq+1) \leq \log_2(p+1)\log_2(q+1), \tag{2}$$
$$\log_2\left(1 + \frac{1}{pq}\right) \leq \log_2\left(1 + \frac{1}{p}\right)\log_2\left(1 + \frac{1}{q}\right). \tag{3}$$

For fixed $p \geq 1$, define
$$F_p(q) = \frac{\log(pq+1)}{\log(q+1)} \qquad (q > 1),$$
where log is the natural logarithm. The derivative has the sign of
$$p(q+1)\log(q+1) - (pq+1)\log(pq+1).$$
With $u = q+1$ and $v = pq + 1 = 1 + p(u-1)$, this sign is non-positive by Lemma 1. Indeed, $v \geq u$, and hence
$$\frac{v \log v}{v-1} \geq \frac{u \log u}{u-1}.$$
Since $v - 1 = p(u-1)$, this gives $v \log v \geq pu \log u$, which is exactly the required non-positivity of the derivative sign. Hence F_p is decreasing, and $F_p(q) \leq F_p(1) = \log(p+1)/\log 2$. This is (2).

For (3), write $a = 1/p$ and $b = 1/q$. For fixed $0 < a \leq 1$, define
$$G_a(b) = \frac{\log(1+ab)}{\log(1+b)} \qquad (0 < b \leq 1).$$
The derivative has the sign of
$$a(1+b)\log(1+b) - (1+ab)\log(1+ab).$$
With $u = 1+b$ and $v = 1 + ab = 1 + a(u-1)$, this sign is non-negative by Lemma 1. Here $v \leq u$, and so
$$\frac{v \log v}{v-1} \leq \frac{u \log u}{u-1}.$$
Since $v - 1 = a(u-1)$, this gives $v \log v \leq au \log u$, which is exactly the required non-negativity of the derivative sign. Thus G_a is increasing on $(0, 1]$, and $G_a(b) \leq G_a(1) = \log(1+a)/\log 2$. This is (3).

Taking $p = i+1$ and $q = j+1$ gives the asserted inequalities for B_ℓ and Δ_ℓ. □

2 The Strengthened Scalar Estimate

For $0 < \alpha < 1$, set

$$a_\alpha = \inf_{n\in\mathbb{N}} \frac{\varphi(n+1)-\alpha}{\varphi(n)}.$$

Then $0 < a_\alpha < 1$. Positivity follows from $\varphi(n+1) > 1 > \alpha$ for every $n \geq 1$. For the upper bound, use $\varphi(n+1)-\varphi(n) \to 0$: choose n with $\varphi(n+1)-\varphi(n) < \alpha$. Then

$$\frac{\varphi(n+1)-\alpha}{\varphi(n)} < 1,$$

and hence $a_\alpha < 1$. Equivalently,

$$\varphi(n)a_\alpha + \alpha \leq \varphi(n+1) \qquad (n \in \mathbb{N}). \tag{4}$$

For $u \geq 0$, define

$$M_\alpha(u) = \max\left\{u, \alpha, \sup_{m\in\mathbb{N}} \frac{\varphi(m)u+\alpha}{\varphi(m+1)}\right\}.$$

If $0 \leq u < a_\alpha$, then $M_\alpha(u) \leq 1$: indeed $u < 1$ and $\alpha < 1$, while (4) gives

$$\frac{\varphi(m)u+\alpha}{\varphi(m+1)} \leq \frac{\varphi(m)a_\alpha+\alpha}{\varphi(m+1)} \leq 1 \qquad (m \in \mathbb{N}).$$

Lemma 3 (strengthened scalar estimate)**.** *Let $0 < \alpha < 1$, $0 \leq s \leq t$, and $k \geq 2$. Then*

$$M_\alpha(s) + \min\{\varphi(k-1)t, \varphi(k)t - s\} \leq \varphi(k)M_\alpha(t). \tag{5}$$

Proof. Write

$$a = \varphi(k-1), \qquad b = \varphi(k), \qquad \delta = b - a.$$

Since

$$\delta = \log_2(k+1) - \log_2(k) = \log_2\left(1+\frac{1}{k}\right),$$

we have $0 < \delta < 1$, and therefore $\delta t \in [0,t]$. It suffices to prove (5) for each affine term in the maximum defining $M_\alpha(s)$.

For the term s, use the second member of the minimum:

$$s + \min\{at, bt-s\} \leq bt \leq bM_\alpha(t).$$

For the term α, use the first member of the minimum and the $(k-1)$-st affine term in $M_\alpha(t)$:

$$\alpha + \min\{at, bt-s\} \leq \alpha + at \leq bM_\alpha(t).$$

It remains to consider the m-th affine term. Put

$$c = \varphi(m), \qquad d = \varphi(m+1), \qquad \eta = d - c.$$

For fixed t, the function

$$s \mapsto \frac{cs+\alpha}{d} + \min\{at, bt-s\}$$

increases on $[0, \delta t]$ and decreases on $[\delta t, t]$. Its maximum over $0 \le s \le t$ is therefore attained at $s = \delta t$, where it equals

$$\frac{\alpha + t(c\delta + ad)}{d} = \frac{\alpha + t(bd - \delta\eta)}{d}.$$

Thus it is enough to prove

$$M_\alpha(t) \ge \frac{\alpha + t(bd - \delta\eta)}{bd}. \tag{6}$$

Apply Lemma 2 with $i = k - 1$ and $j = m$. There is $\ell \in \mathbb{N}$ such that, with

$$D = B_\ell = \varphi(\ell + 1), \qquad \lambda = \Delta_\ell = \varphi(\ell + 1) - \varphi(\ell),$$

we have $D \le bd$ and $\lambda \le \delta\eta$. The ℓ-th affine term in $M_\alpha(t)$ gives

$$M_\alpha(t) \ge \frac{\varphi(\ell)t + \alpha}{\varphi(\ell + 1)} = t + \frac{\alpha - \lambda t}{D}.$$

If $\alpha - \delta\eta t < 0$, the right-hand side of (6) is below t, while $M_\alpha(t) \ge t$. If $\alpha - \delta\eta t \ge 0$, then $\alpha - \lambda t \ge \alpha - \delta\eta t \ge 0$ and $D \le bd$, so

$$M_\alpha(t) \ge t + \frac{\alpha - \delta\eta t}{bd} = \frac{\alpha + t(bd - \delta\eta)}{bd}.$$

This proves (6), hence (5). $\square$

Remark 4. *The factor $M_\alpha(t)$ on the right-hand side of Lemma 3 is what makes the estimate stable under the recursive definition of the Schlumprecht norm.*

3 The Far-Coordinate Perturbation

Define the iterative norms by

$$\|x\|_0 = \|x\|_\infty, \qquad \|x\|_{r+1} = \max\left\{ \|x\|_\infty, \sup_{k \ge 2} \sup_{E_1 < \cdots < E_k} \frac{1}{\varphi(k)} \sum_{i=1}^{k} \|E_i x\|_r \right\}.$$

Then $\|x\|_{\mathcal{S}} = \sup_r \|x\|_r$.

Lemma 5 (comparison by local majorants)**.** *Let $y \in c_{00}$. Suppose that to every finite $F \subset \mathbb{N}$ one assigns $\rho(F) \ge 0$ such that*

$$\begin{aligned} \|Fy\|_\infty &\le \rho(F), \\ G \subset F &\Longrightarrow \rho(G) \le \rho(F), \\ \sum_{i=1}^{k} \rho(E_i) &\le \varphi(k)\rho(F) \end{aligned}$$

whenever $k \ge 2$ and $E_1 < \cdots < E_k$ are finite subsets of F. Then

$$\|Fy\|_{\mathcal{S}} \le \rho(F) \qquad (F \subset \mathbb{N} \textit{ finite}).$$

Proof. Induct on the iterative norms. The assertion is true for $\|\cdot\|_0$ by the first assumption. Suppose it is true for $\|\cdot\|_r$. Fix F and an admissible family $E_1 < \cdots < E_k$, $k \geq 2$. Put $G_i = E_i \cap F$ and discard empty intersections.

If no G_i remains, the sum is zero. If exactly one set G remains, then

$$\|Gy\|_r \leq \rho(G) \leq \rho(F) \leq \varphi(k)\rho(F).$$

The last inequality uses $k \geq 2$, hence $\varphi(k) \geq 1$. If $q \geq 2$ sets $G_1 < \cdots < G_q$ remain, then $q \leq k$, and

$$\sum_{j=1}^{q} \|G_j y\|_r \leq \sum_{j=1}^{q} \rho(G_j) \leq \varphi(q)\rho(F) \leq \varphi(k)\rho(F).$$

Thus every admissible sum in the definition of $\|Fy\|_{r+1}$ is bounded by $\varphi(k)\rho(F)$, and so $\|Fy\|_{r+1} \leq \rho(F)$. Taking the supremum over r proves the claim. $\square$

Lemma 6 (far-coordinate unit-ball lemma)**.** *Let $0 < \alpha < 1$. If $x \in c_{00}$ and $\|x\|_{\mathcal{S}} < a_\alpha$, then, for every $N > \max \operatorname{supp} x$,*

$$\|x + \alpha e_N\|_{\mathcal{S}} \leq 1, \qquad \|x - \alpha e_N\|_{\mathcal{S}} \leq 1.$$

Proof. By 1-unconditionality and $N \notin \operatorname{supp} x$,

$$\|x + \alpha e_N\|_{\mathcal{S}} = \||x| + \alpha e_N\|_{\mathcal{S}}, \qquad \|x - \alpha e_N\|_{\mathcal{S}} = \||x| + \alpha e_N\|_{\mathcal{S}}.$$

Thus it is enough to consider $x \geq 0$ and $y = x + \alpha e_N$. For each finite $F \subset \mathbb{N}$, define

$$\rho(F) = \begin{cases} \|Fx\|_{\mathcal{S}}, & N \notin F, \\ M_\alpha(\|Fx\|_{\mathcal{S}}), & N \in F. \end{cases}$$

The lattice monotonicity of the Schlumprecht norm and the monotonicity of M_α give $G \subset F \Rightarrow \rho(G) \leq \rho(F)$. Also $\|Fy\|_\infty \leq \rho(F)$, since $M_\alpha(u) \geq u$ and $M_\alpha(u) \geq \alpha$.

We verify the local majorant inequality. Fix finite F and $E_1 < \cdots < E_k \subset F$, $k \geq 2$. If $N \notin F$, then

$$\sum_i \rho(E_i) = \sum_i \|E_i x\|_{\mathcal{S}} \leq \varphi(k)\|Fx\|_{\mathcal{S}} = \varphi(k)\rho(F).$$

Assume $N \in F$ and set $t = \|Fx\|_{\mathcal{S}}$. If none of the E_i contains N, then

$$\sum_i \rho(E_i) = \sum_i \|E_i x\|_{\mathcal{S}} \leq \varphi(k)t \leq \varphi(k)M_\alpha(t) = \varphi(k)\rho(F).$$

If exactly one set, say E_j, contains N, put $s = \|E_j x\|_{\mathcal{S}}$. The remaining pieces satisfy

$$\sum_{i \neq j} \|E_i x\|_{\mathcal{S}} \leq \varphi(k-1)t, \qquad \sum_{i \neq j} \|E_i x\|_{\mathcal{S}} \leq \varphi(k)t - s.$$

For the first inequality, if $k = 2$ then there is only one remaining piece, say E_i, and

$$\sum_{i \neq j} \|E_i x\|_{\mathcal{S}} = \|E_i x\|_{\mathcal{S}} \leq \|Fx\|_{\mathcal{S}} = t = \varphi(1)t.$$

If $k > 2$, the $k-1$ remaining pieces form an ordered admissible family, so the norm recursion gives the bound with $\varphi(k-1)$. The second inequality follows from the full k-tuple. Lemma 3 gives

$$\sum_i \rho(E_i) \leq M_\alpha(s) + \min\{\varphi(k-1)t, \varphi(k)t - s\} \leq \varphi(k)M_\alpha(t) = \varphi(k)\rho(F).$$

Lemma 5 therefore yields $\|Fy\|_{\mathcal{S}} \le \rho(F)$ for every finite F. Taking $F = \operatorname{supp} x \cup \{N\}$ gives

$$\|x + \alpha e_N\|_{\mathcal{S}} \le M_\alpha(\|x\|_{\mathcal{S}}) \le 1,$$

because $\|x\|_{\mathcal{S}} < a_\alpha$. The estimate for $x - \alpha e_N$ follows from the unconditionality reduction at the start of the proof. □

4 The Converse Formula

For a Banach space X, use the source paper's notation

$$r_X(\varepsilon) = \sup\{r > 0 : rB_{X^*} \subset s_\varepsilon(B_{X^*})\}.$$

For $X = \mathcal{S}^*$, identify $X^* = \mathcal{S}^{**}$ with $\mathcal{S}$, since $\mathcal{S}$ is reflexive.

Theorem 7. *For every* $0 < \varepsilon < 2$*,*

$$r_{\mathcal{S}^*}(\varepsilon) = \inf_{n\in\mathbb{N}} \frac{\log_2(n+2) - \varepsilon/2}{\log_2(n+1)}.$$

Proof. Let $\beta = \varepsilon/2$ and

$$a_\beta = \inf_{n\in\mathbb{N}} \frac{\varphi(n+1) - \beta}{\varphi(n)}.$$

The upper bound $r_{\mathcal{S}^*}(\varepsilon) \le a_\beta$ is (1), proved by Kochanek and Miarka. We prove the reverse inequality.

First note that $\alpha \mapsto a_\alpha$ is continuous on $(0,1)$: it is the infimum of affine functions with slopes bounded in absolute value by 1. Let $x \in c_{00}$ and $\|x\|_{\mathcal{S}} < a_\beta$. Choose $\alpha \in (\beta, 1)$ close enough to β that $\|x\|_{\mathcal{S}} < a_\alpha$. For every $N > \max \operatorname{supp} x$, Lemma 6 gives

$$x + \alpha e_N,\ x - \alpha e_N \in B_{\mathcal{S}}.$$

The basis of $\mathcal{S}$ is weakly null, because $\mathcal{S}$ is reflexive and its basis is shrinking. Hence both sequences $x + \alpha e_N$ and $x - \alpha e_N$ converge weakly, equivalently weak-star in $\mathcal{S} = (\mathcal{S}^*)^*$, to x. Their mutual distance is

$$\|(x + \alpha e_N) - (x - \alpha e_N)\|_{\mathcal{S}} = 2\alpha > \varepsilon.$$

Thus every weak-star neighborhood of x intersects $B_{\mathcal{S}}$ in a set of diameter greater than ε, and $x \in s_\varepsilon(B_{\mathcal{S}})$.

Now fix $0 < r < a_\beta$ and $z \in rB_{\mathcal{S}}$. Choose $z_m \in c_{00}$ with $z_m \to z$ in norm. For all large m, $\|z_m\|_{\mathcal{S}} < a_\beta$, hence $z_m \in s_\varepsilon(B_{\mathcal{S}})$. The Szlenk derivation of the weak-star compact ball is weak-star closed, hence norm closed. Indeed, if K is weak-star compact and $x \in K \setminus s_\varepsilon(K)$, then some weak-star open neighborhood $V \ni x$ has $\operatorname{diam}(K \cap V) \le \varepsilon$; the same V shows every $y \in K \cap V$ also lies outside $s_\varepsilon(K)$, so $K \setminus s_\varepsilon(K)$ is relatively weak-star open. Therefore $z \in s_\varepsilon(B_{\mathcal{S}})$. Since this holds for every $z \in rB_{\mathcal{S}}$ and every $r < a_\beta$, we have $r_{\mathcal{S}^*}(\varepsilon) \ge a_\beta$. Combined with the source paper's upper estimate, this proves the formula. □

Verification Notes

The packet includes `code/check_scalar_and_finite_cases.py`. The scalar estimate is proved by Lemma 2 and Lemma 3; the code is only a sanity check. It tests the scalar inequalities on a large finite grid and random sample and computes representative finite-support Schlumprecht norms by interval dynamic programming. The command used was:

```
conda run --no-capture-output -n sandbox python
code/check_scalar_and_finite_cases.py
```

Novelty and Literature Check

Bounded check performed on June 14, 2026: searched the arXiv page for arXiv:2409.05516 and web queries for the exact source title together with "Schlumprecht", "Szlenk", "converse", and "r_{S^*}". This found the source paper and no later paper explicitly resolving the converse formula. This is not an exhaustive bibliographic search, but no immediate literature-already-answered match was found.

Human Review Recommendation

This should be reviewed as a candidate full solution. The most important points to check are the product lemma, the strengthened scalar estimate, and the local majorant comparison lemma.

References

[1] T. Kochanek and M. Miarka, *When is the Szlenk derivation of a dual unit ball another ball?*, arXiv:2409.05516 (2024).

[2] T. Schlumprecht, *An arbitrarily distortable Banach space*, Israel J. Math. 76 (1991), 81–95.